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On multiscale epidemic models

Fernando Saldaña

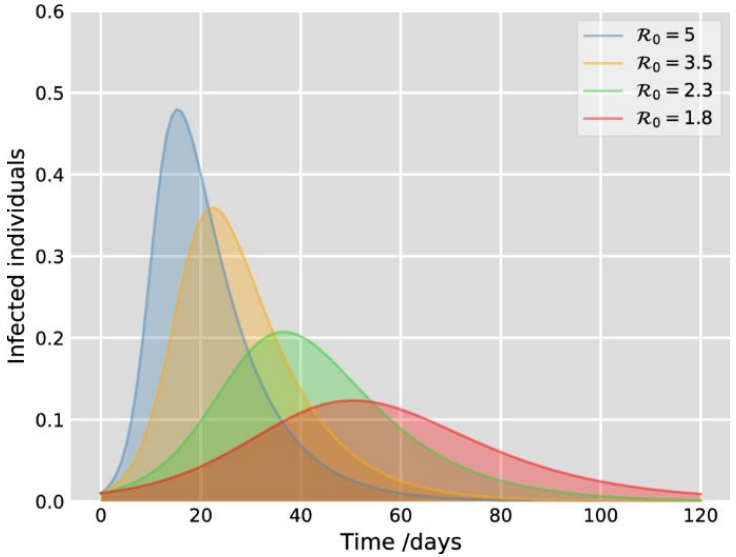
Universidad Complutense de Madrid

Seminario AMMA

22 de Octubre 2026

Talk structure

- I: Introduction: Why epidemic modelling?
- II: Multiscale epidemic modelling
- III: Optimal control for HPV vaccination (optional)



Taken from
Saldaña, F., Velasco-Hernández, J.X. Modeling the COVID-19 pandemic: a primer and overview of mathematical epidemiology. *SeMA* **79**, 225–251 (2022).



I: Introduction
Why epidemic modelling?

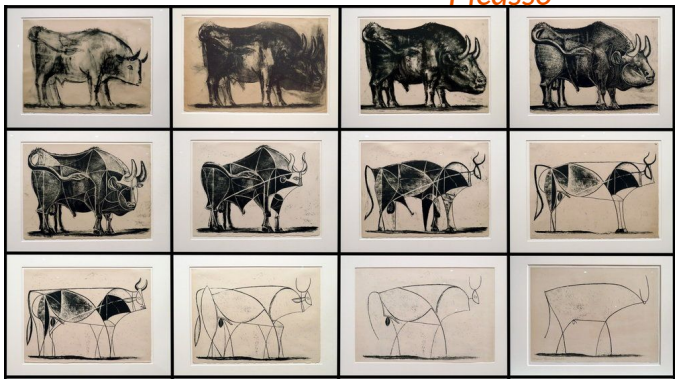
Why Mathematical Models Matter in Epidemic Analysis

We can answer the following questions:

- How many people need to be vaccinated to stop transmission?
- What is the risk posed by a new virus?
- How fast will an infection spread?
- What are the expected future case numbers? (Hospital demand)
- When will the peak occur?
- How long will an epidemic last?

What is a model?

Picasso

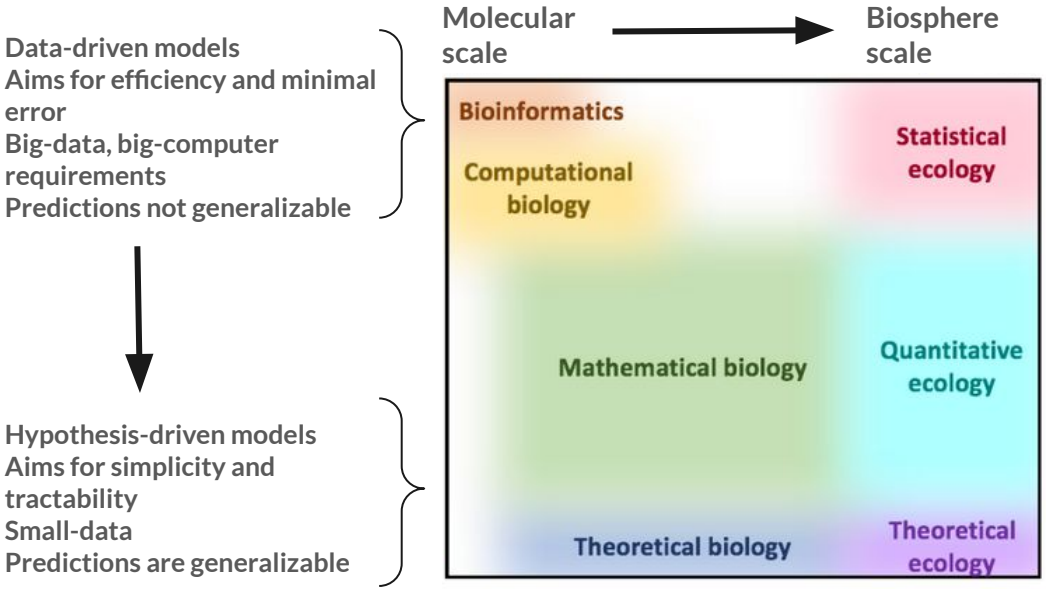


Models are simplified descriptions of a process

"All models are wrong, but some are useful" George Box

Data-driven models
Aims for efficiency and minimal error
Big-data, big-computer requirements
Predictions not generalizable

Hypothesis-driven models
Aims for simplicity and tractability
Small-data
Predictions are generalizable

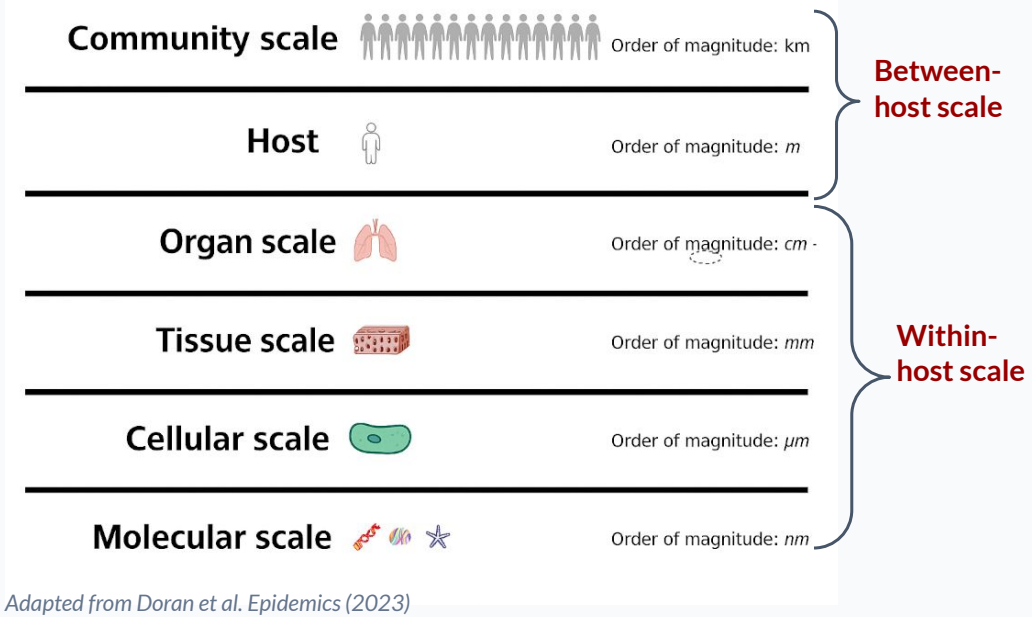


Credits: Rob Noble

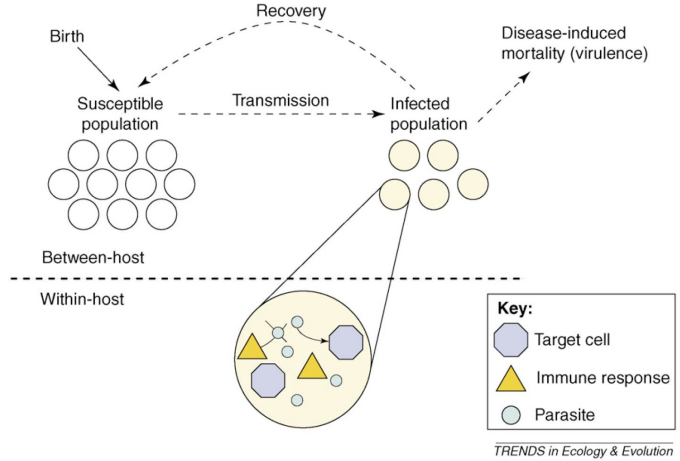


II: Multiscale epidemic modeling

Overview: Infectious disease systems are multiscale processes



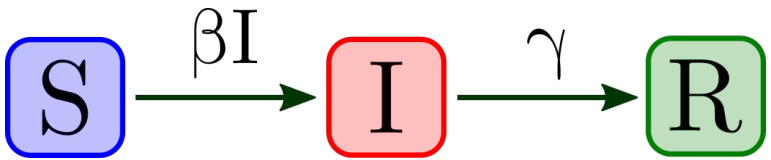
Adapted from Doran et al. Epidemics (2023)



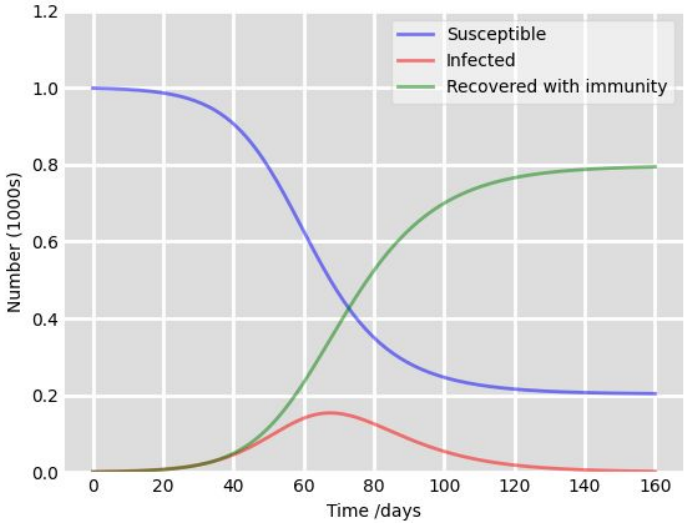
Taken from Mideo, Alizon, and Day. Cell (2008)

Infectious diseases spread in a complex manner involving multiple host and various temporal and spatial scales. These scales can be broadly grouped into two scales: i) the within-host scale and ii) the between-host scale.

The between-host scale: The SIR model



S = susceptible
I = Infectado
R = Recuperado



$$\frac{dS}{dt} = -\beta SI$$

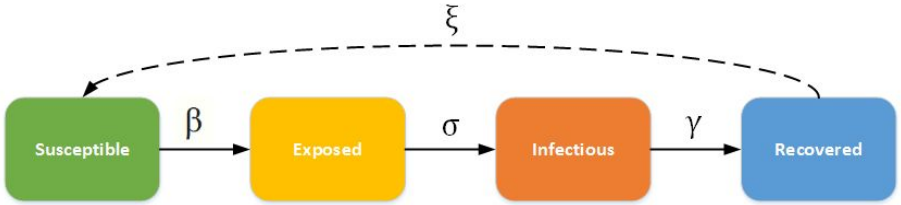
$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

The SEIRS model

The SEIRS model extends SIR including:

1. **Exposed class: infected but not infectious**
2. Loss of immunity



$$\frac{dS}{dt} = -\beta SI + \varepsilon R,$$

$$\frac{dE}{dt} = \beta SI - \sigma E,$$

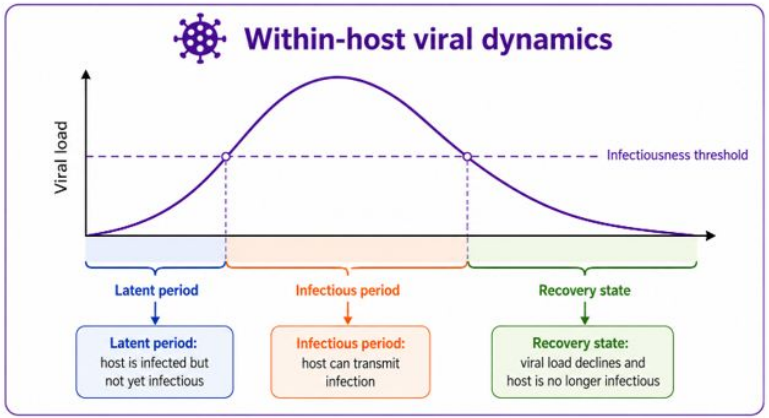
$$\frac{dI}{dt} = \sigma E - \gamma I,$$

$$\frac{dR}{dt} = \gamma I - \varepsilon R.$$

Why linking scales is important?

Within-host dynamics influence:

- Probability of infection
- Latent and infectious period
- Disease-induced mortality



Challenges in multiscale epidemic models

- Lack of data to calibrate models
- No standard modeling framework to link scale

$$\beta(V) = cV?$$

Relationship pathogen load - infectiousness has not been determined for most pathogens

- Difficult mathematical analysis and high computational costs

Objective & Methods

Objective

Evaluate when is it necessary to account for within-host dynamics, and when are single-scale models sufficient to capture epidemiological behavior?

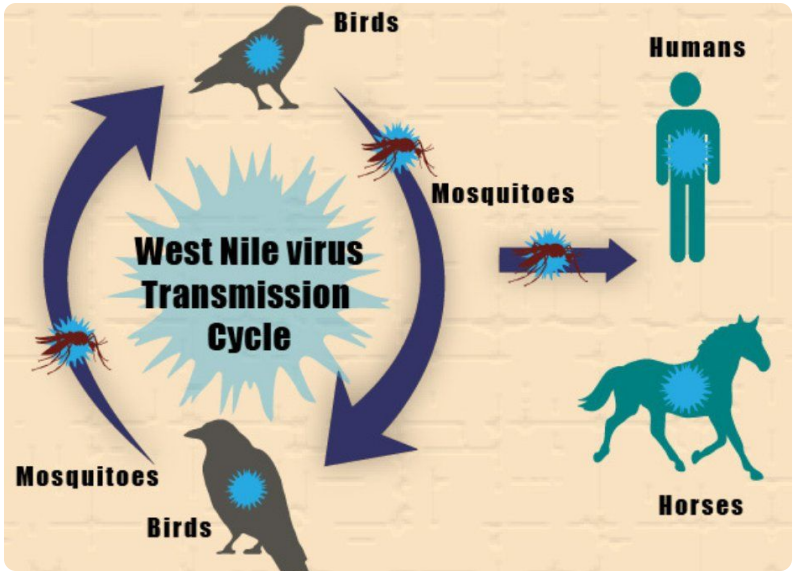
Methods

Comparison of a multiscale model vs an “equivalent” single-scale model

Application

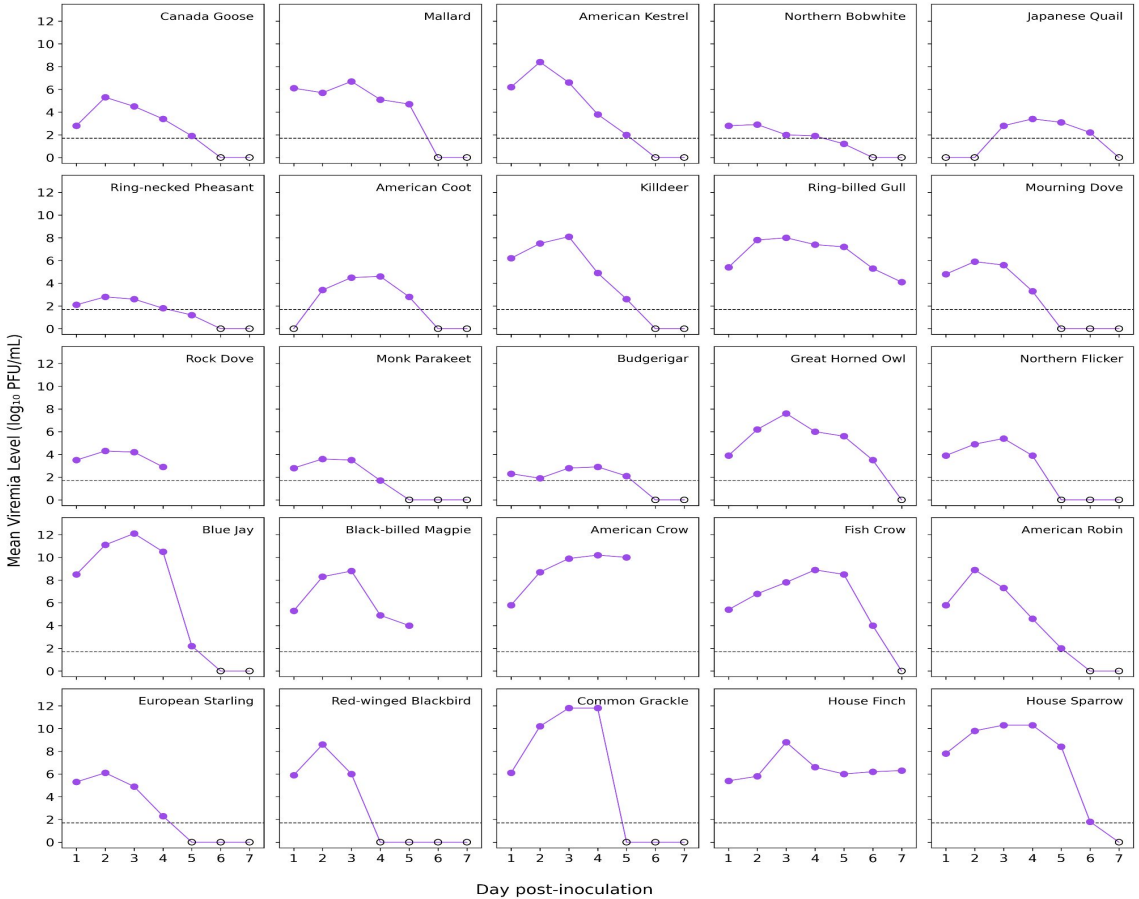
WNV flavivirus transmitted by mosquitoes. Avian hosts are the natural reservoir. Humans are incidental hosts.

Case of study: West Nile Virus (WNV)



WNV in birds

Bird species differ greatly in their **viral profile**, which leads to differences in their capacity to infect, duration of the infectious period, mortality, etc.



Experimental data from 25 bird species infected with WNV. From Komar et al. 2003

Multiscale model formulation

Stochastic individual based model:
Dynamic bipartite network.
Nodes represent hosts and mosquitoes.
Edges = bites (they change each time step)

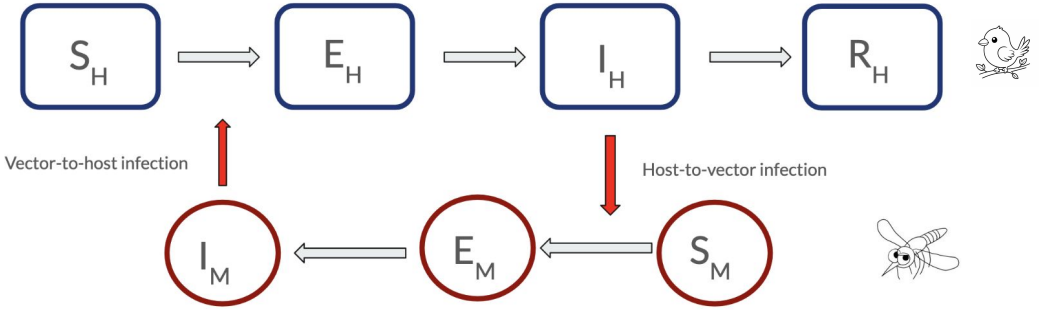
Mosquito bites: Blood meal every 4 days -> Poisson distribution with mean 0.25

Within-host dynamics: unique viral profile for each host based on a phenomenological model

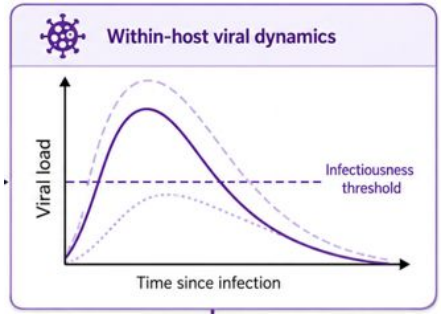
Linking of scales: Viral load determines infectious status + dose-response model to determine probability of infection as function of host viral load

Mosquito EIP: Extrinsic incubation period follows a Weibull distribution

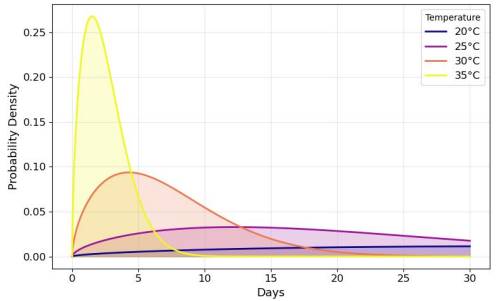
Between-host: SEIR+D avian hosts + SEI mosquitoes



Within-host: Unique viral profiles



Within-vector: Weibull distribution for the EIP



Within-host model and parameter estimation

Phenomenological model:

$$V(t) = \frac{2V_p}{e^{-l_g(t-T_p)} + e^{l_d(t-T_p)}}$$

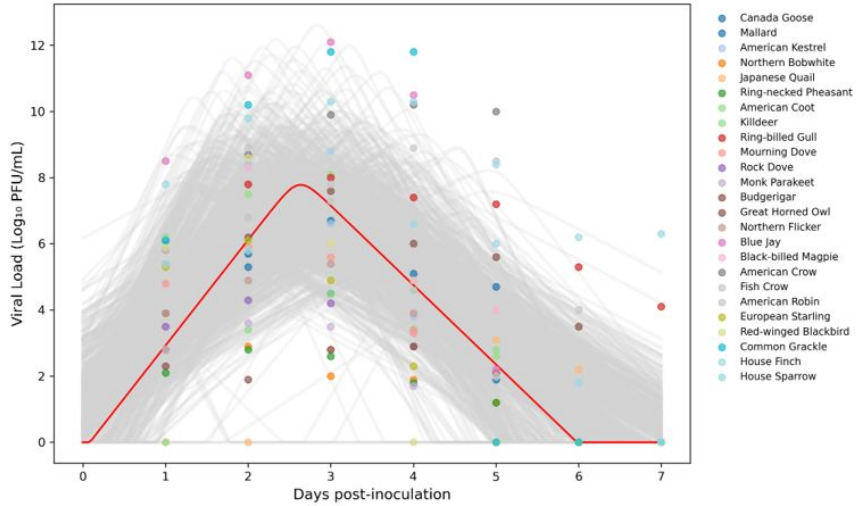
V(t) = Viral load

Vp = Viral peak

Tp = Time of viral peak

l_g = Exponential growth rate

l_d = Exponential decline rate



Parameter estimation:
Point estimate -> MLE
Uncertainty -> Parametric bootstrapping (synthetic datasets with noise) -> empirical CIs

Viremia-infectiousness relationship

Dose-response model for probability of mosquito infection as a function of host viral load

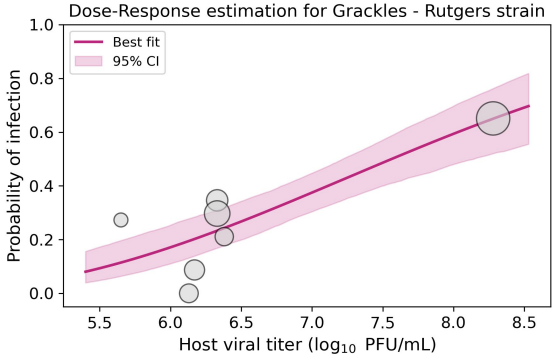
$$p\left(V(\tau);a,h\right)=\begin{cases}1-\exp\left[-a\left(\log_{10}V(\tau)-\log_{10}D\right)^{h}\right], & V(\tau)>D\\ 0, & V(\tau)\leq D\end{cases}$$

Parameters to be estimated: “a”, y “h”

D = Minimum viral load required for transmission (10^4 PFU/ml for WNV)

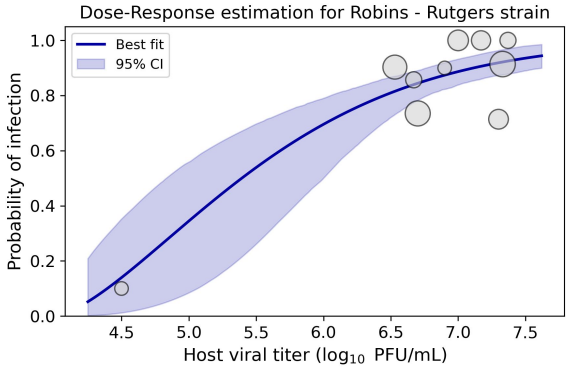
Zanate





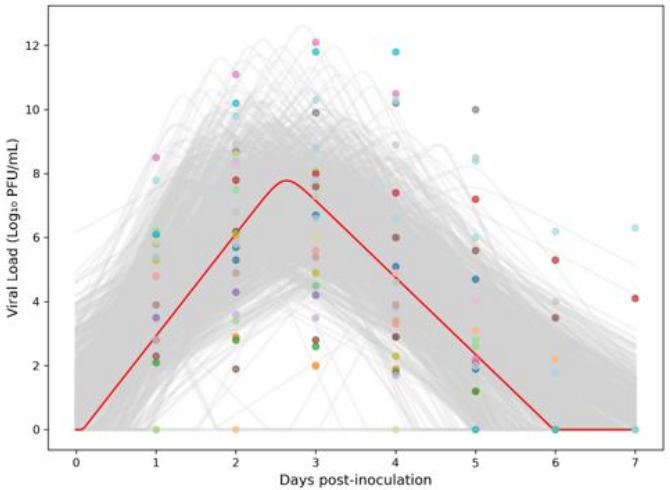
Robin

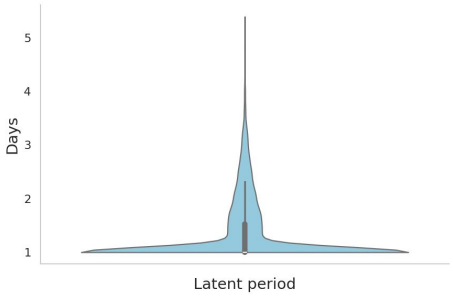


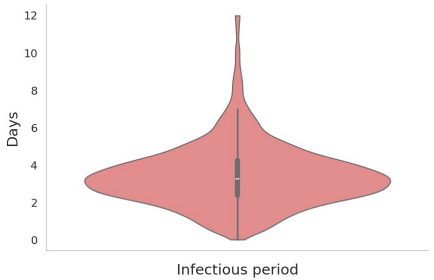


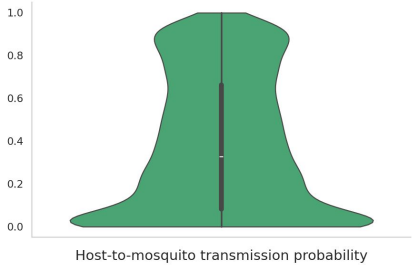
Single-scale “equivalent” model

“equivalence” means that the single-scale model is calibrated to reproduce the same average epidemiological traits generated by the multiscale model. All other assumptions remain the same











Violin plots for the distribution of the latent period, infectious period, transmission probability and disease-induced mortality rate. Mean values are used in the single-scale model

Comparison methods

Comparison based on 300 realizations for differents vector-to-host ratios (VHR)

Metrics of comparison includes:

- R0
- Peak host prevalence
- Final epidemic size (fraction)
- Time to peak prevalence
- Epidemic growth rate
- Mean and max mosquitoes infected per host

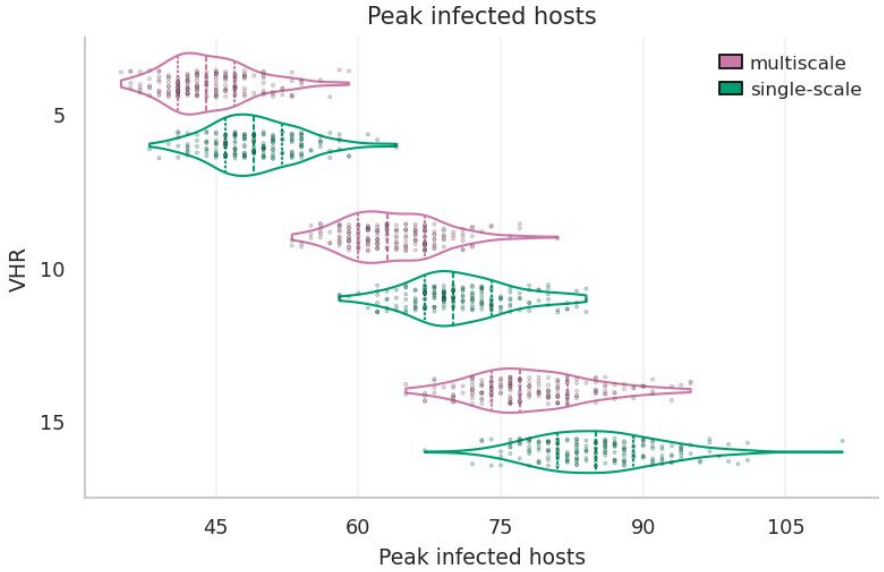
Statistical comparison of model outcomes for each metric

- Comparison of central tendencies using medians
- Comparison of empirical distributions using Wasserstein distance
- p-values derived using a two-sided permutation test

Results (Peak infected hosts)

- For all VHRs, difference in medians around 10% (statistically significant)
- Significant difference in distributions

Peak infected hosts is different across formulations



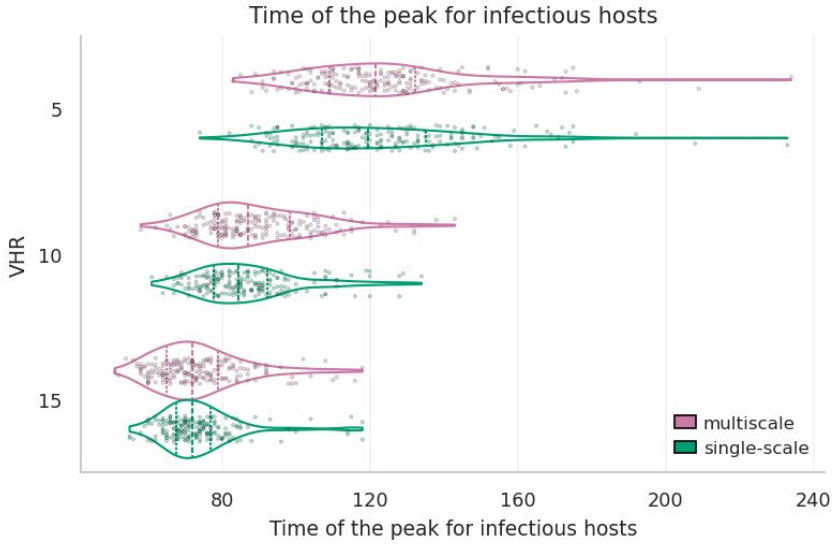
Violin plot for the distribution of the peak infected host. Internal lines show the quartiles including the median.

Results (Peak time)

- For all VHRs, difference in medians below 3% (non significant)
- Non-significant difference in distributions

Peak time is equivalent across formulations

Epidemic doubling time is also equivalent (not shown)



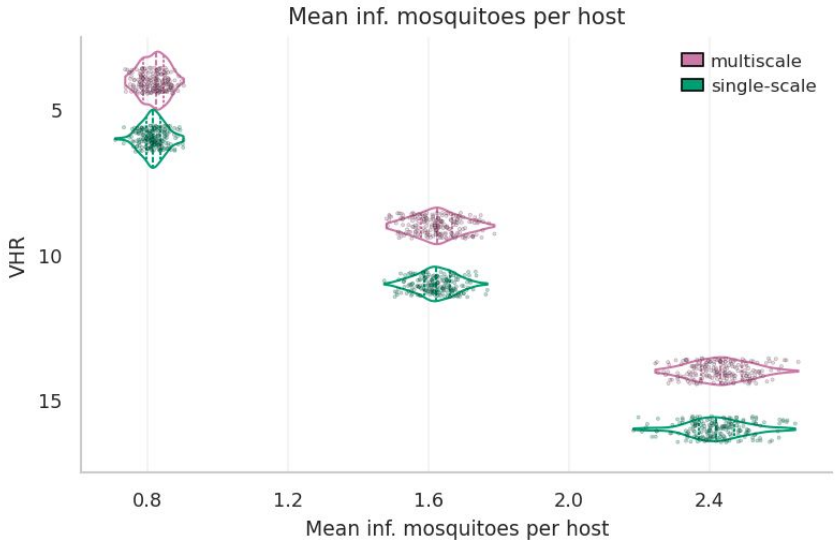
Violin plot for the distribution of the peak infected host. Internal lines show the quartiles including the median.

Results (mean mosq. inf. per host)

- For all VHRs, difference in medians below 2% (not significant)
- Non significant difference in distributions

Mean inf. mosquitoes infected per host is equivalent across formulations

Max. inf. mosquitoes per host is also equivalent (not shown)

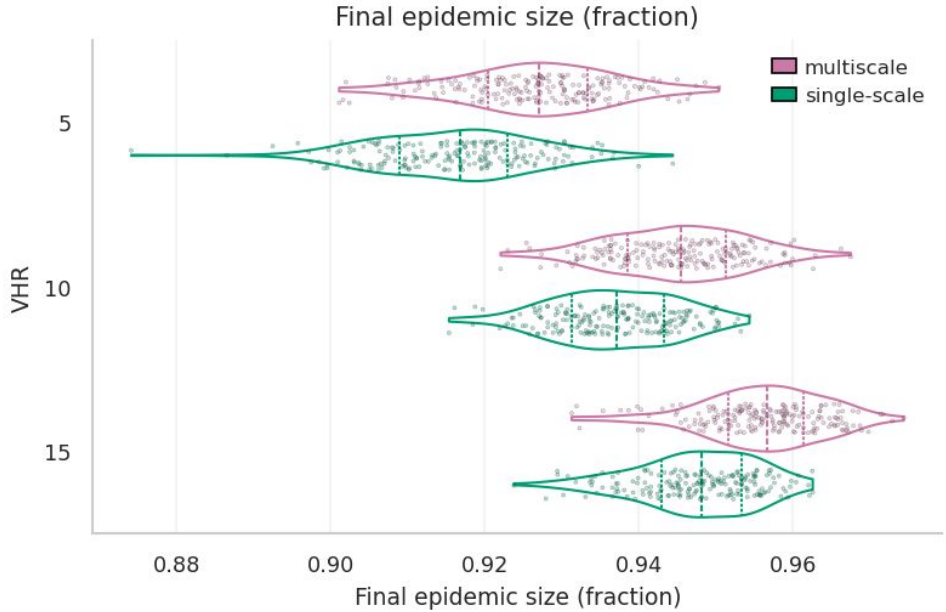


Violin plot for the distribution of the mean inf. mosquitoes per host. Internal lines show the quartiles including the median.

Results (Final epidemic size)

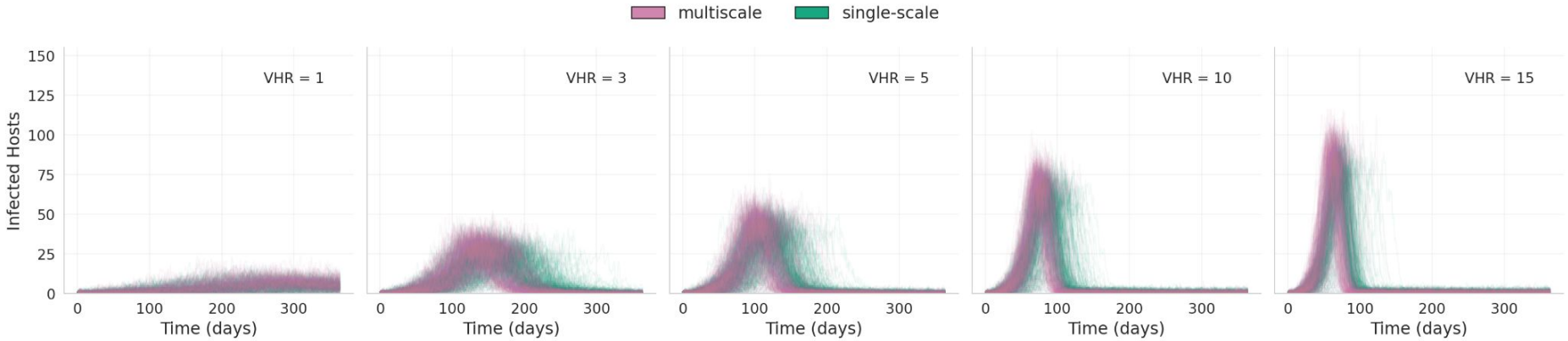
- For all VHRs, difference in medians in below 1.2% (no statistical difference)
- Difference in distribution is only significant for VHR=5

Final epidemic size is equivalent for both formulations (for high VHR).



Violin plot for the distribution of the final epidemic size. Internal lines show the quartiles including the median.

Results (time series infected hosts)



Time series for the infectious host class based on an ensemble of 300 stochastic realizations.

Conclusion (until now)

In many contexts, single-scale models provide a good representation of epidemiological dynamics yet they should be properly calibrated to correctly represent within-host dynamics

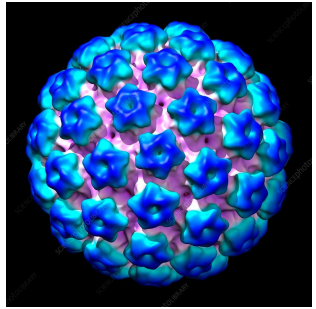
Future work

- Include immunological dynamics in the within-host model
- Include realistic seasonality based on temperature
- Include mosquito feeding preferences
- Consider spatial structure

III: Case of study
HPV vaccine optimization

What is HPV? Key facts

- Human papillomavirus (HPV) infection is the most common sexually transmitted infection
- 80% of people get the infection at some point
- **HPV is the main responsible for cervical cancer**
- HPV also causes anal, penile, oropharyngeal cancers
- HPV vaccines are highly effective to prevent infection



HPV vaccination strategies: Female-only vs Gender-Neutral (GN)

Many programs focus on girls only, based on:

- Cervical Cancer (CC) being the most important HPV burden
- high coverage in girls provides herd-immunity in boys

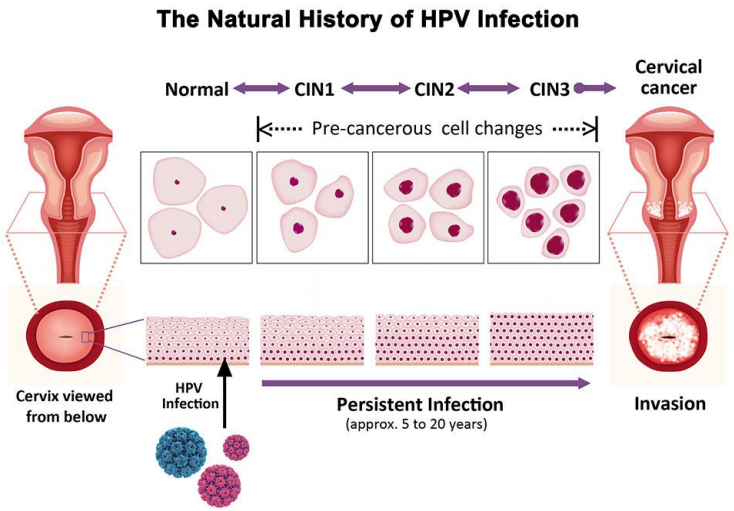
Why include boys in HPV vaccination?

- Direct protection for men (anal, penile, oropharyngeal cancers)
- Protection for men who have sex with men
- Herd immunity is easier to achieve including men
- Greater health equity

Under what conditions the inclusion of boys in HPV vaccination programs is cost-effective?

Two-sex HPV epidemic model with vaccination

S = Susceptible
V = Vaccinated
A = Asymptomatic infectious
I = Infectious with persistent infection
P = People in precancerous stage
C = People with cancer



ODE Model (12 equations)

$$\dot{S}_j = (1-w_j)\,\Lambda_j - (\lambda_i + u_j + \mu_j)\,S_j + q_A^j k_A^j A_j + q_I^j k_I^j I_j + q_P^j k_P^j P_j + \theta_j V_j,$$

$$\dot{V}_j = w_j \Lambda_j + u_j S_j - (\mu_j + \theta_j)\,V_j,$$

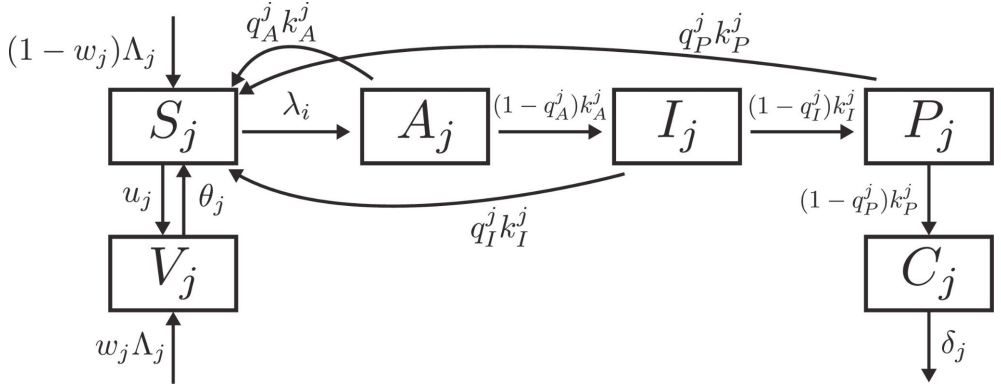
$$\dot{A}_j = \lambda_i S_j - \left(k_A^j + \mu_j\right) A_j,$$

$$\dot{I}_j = \left(1 - q_A^j\right) k_A^j A_j - \left(k_I^j + \mu_j\right) I_j,$$

$$\dot{P}_j = \left(1 - q_I^j\right) k_I^j I_j - \left(k_P^j + \mu_j\right) P_j,$$

$$\dot{C}_j = \left(1 - q_P^j\right) k_P^j P_j - (\delta_j + \mu_j)\,C_j,$$

Flow diagram of the model



Results

New vaccination programs:

- If the goal is to reduce all HPV-associated diseases, vaccination should include both sexes
- Even if the objective is to reduce CC, beginning with GN vaccination helps rapidly lower prevalence; once uptake is high, focus only in females

Existing girls-only programs:

- With >30% coverage in females and 0% in males, increase coverage in females is the most cost-effective
- However, GN vaccination still achieves the lowest prevalence thus, if resources allow, universal vaccination is preferable

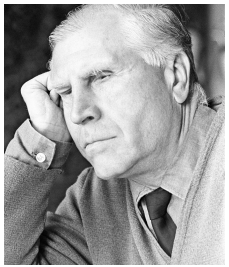
Methods: Optimal control theory (OCT)

OCT is a branch of control theory that deals with finding a control for a dynamical system:

System: (e.g. epidemic model)
Control action: (e.g. vaccination rate)
Goal: Find the control strategy that optimizes a chosen objective, such as: minimizing cervical cancer cases.

Math: Proof of existence & unicity, find optimality system, FBSM

"Pontryagin's Maximum Principle is one of the main theoretical results in OCT"



Pontryagin

Thank you for your time

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